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01

10 questions · ~15 min

Partial List of Candidate Species for De-extinction

Common name	Scientific name	Became extinct
Huia	<i>Heteralocha acutirostris</i>	1907
Caribbean monk seal	<i>Monachus tropicalis</i>	1952
Passenger pigeon	<i>Ectopistes migratorius</i>	1914
Saber-toothed cat	<i>Smilodon</i>	11,000 years before present
Woolly mammoth	<i>Mammuthus primigenius</i>	6,400 years before present

The passage of time is among the many obstacles faced by scientists who are pursuing de-extinction efforts—that is, efforts to use breeding or a mixture of cloning and genetic engineering to bring back extinct species. Specifically, researchers are concerned that the longer a species has been extinct, the less likely it is that a suitable habitat still exists for that species. Among candidate species for de-extinction, this problem would be especially concerning for the _____blank

Which choice most effectively uses data from the table to complete the statement?

- A. passenger pigeon (*Ectopistes migratorius*), which became extinct only a few years after the huia (*Heteralocha acutirostris*).
- B. saber-toothed cat (*Smilodon*), which became extinct 11,000 years ago.

- C. woolly mammoth (*Mammuthus primigenius*), which became extinct several thousand years before the saber-toothed cat (*Smilodon*).
- D. Caribbean monk seal (*Monachus tropicalis*), which became extinct in 1952.

The ancient writing system used in the Maya kingdoms of southern Mexico and Central America had a symbol for the number zero. The earliest known example of the symbol dates to more than 2,000 years ago. At that time, almost none of the writing systems elsewhere in the world possessed a zero symbol. And the use of zero in Mexico and Central America may be even more ancient. Some historians suggest that Maya mathematicians inherited it from the Olmec civilization, which flourished in the region 2,400–3,600 years ago.

According to the text, what do some historians suggest about Maya civilization?

- A. Maya civilization acquired the use of zero from the Olmec civilization.
- B. Maya civilization respected its historians more than it respected its mathematicians.
- C. Maya civilization was highly secretive about its intellectual achievements.
- D. Maya civilization tried to introduce its writing system to other civilizations.

The Household, Income and Labour Dynamics in Australia Survey (HILDA) examines trends in economic well-being among 25,000 people in Australia unfolding over many years. As is true of most longitudinal studies, this need for years of data collection results in high costs. By contrast, a relatively straightforward fitness study, such as one that is merely trying to identify the percentage of regular exercisers in a city who do weight training, may not need a large budget because _____ blank

Which choice most logically completes the text?

- A. longitudinal methods are probably suitable for the fitness study.
- B. it would be easy for HILDA researchers to add questions to their economic well-being study.
- C. 25,000 people is more than enough for HILDA to find trends in economic well-being.
- D. the fitness study can be done well without years of data collection.

Jan Gimsa, Robert Sleigh, and Ulrike Gimsa have hypothesized that the sail-like structure running down the back of the dinosaur *Spinosaurus aegyptiacus* improved the animal's success in underwater pursuits of prey species capable of making quick, evasive movements. To evaluate their hypothesis, a second team of researchers constructed two battery-powered mechanical models of *S. aegyptiacus*, one with a sail and one without, and subjected the models to a series of identical tests in a water-filled tank.

Which finding from the model tests, if true, would most strongly support Gimsa and colleagues' hypothesis?

- A. The model with a sail took significantly less time to complete a sharp turn while submerged than the model without a sail did.
- B. The model with a sail displaced significantly more water while submerged than the model without a sail did.
- C. The model with a sail had significantly less battery power remaining after completing the tests than the model without a sail did.
- D. The model with a sail took significantly longer to travel a specified distance while submerged than the model without a sail did.

The painter María Izquierdo played an important role in the development of twentieth-century Mexican art, but her work has never been well-known in the United States except among art historians. One reason for Izquierdo's relative obscurity is the enormous popularity of some of her peers. In particular, the painters Frida Kahlo and Diego Rivera have so captivated the interest of US audiences that Izquierdo and other Mexican artists from the period often get overlooked, despite the high quality of their work.

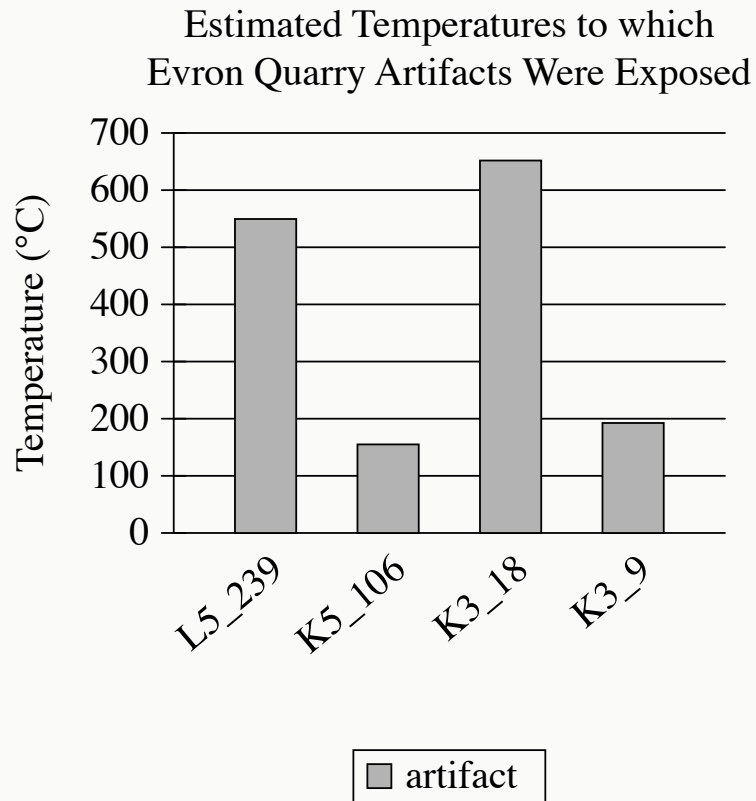
Which choice best states the main idea of the text?

- A. Izquierdo's work is not as well-known in the United States as it should be because Kahlo and Rivera draw so much of the public's attention.
- B. During Izquierdo's lifetime, her paintings were displayed in galleries in the United States much more frequently than paintings by Kahlo and Rivera were.
- C. Izquierdo painted some of the same subjects that Kahlo and Rivera painted but used different techniques than they used.
- D. Few of Izquierdo's works are in galleries today because she produced only a small number of paintings.

In a study of the mechanisms underlying associative memory—or the ability to learn and remember connections between inherently unrelated things—neuroscientists Kei Igarashi, Jasmine Chavez, and others presented mice with memory tests. The team discovered that fan cells, a type of cell found in the medial temporal lobe of the brain, are necessary for the acquisition of new associative memories. They also found that fan cell activity requires dopamine, a chemical the brain produces in response to pleasure and rewards. Consequently, receiving a reward should likely help to _____ blank

Which choice most logically completes the text?

- A. decrease an individual's capacity to utilize dopamine.
- B. increase an individual's capacity to recognize differences between unrelated things.
- C. increase an individual's capacity to form associative memories.
- D. decrease an individual's capacity to create fan cells.



Flint artifacts dating to 800,000 to 1,000,000 years ago have been recovered from the Evron Quarry in Israel. Likely created by the hominin *Homo erectus*, the artifacts have no visual features suggesting that they were exposed to fire, leading some scholars to conclude that these hominins had not acquired control of fire. But Zane Stepka and colleagues recently used a new method to determine whether these artifacts had been exposed to temperatures above 400°C (the typical temperature campfires reach) and concluded that the hominins who inhabited the site may have had control of fire.

Which choice best describes data in the graph that support the team's conclusion?

- A. Artifacts K5_106 and K3_9 were exposed to temperatures above 400°C.

- B. Artifacts L5_239 and K3_18 were exposed to temperatures of approximately 550°C and 650°C, respectively.
- C. All of the artifacts were exposed to temperatures above 100°C.
- D. Artifact K3_9 was exposed to a higher temperature than was artifact K5_106.

To dye wool, Navajo (Diné) weaver Lillie Taylor uses plants and vegetables from Arizona, where she lives. For example, she achieved the deep reds and browns featured in her 2003 rug *In the Path of the Four Seasons* by using Arizona dock roots, drying and grinding them before mixing the powder with water to create a dye bath. To intensify the appearance of certain colors, Taylor also sometimes mixes in clay obtained from nearby soil.

Which choice best states the main idea of the text?

- A. Reds and browns are not commonly featured in most of Taylor's rugs.
- B. Taylor draws on local resources in the approach she uses to dye wool.
- C. Taylor finds it difficult to locate Arizona dock root in the desert.
- D. *In the Path of the Four Seasons* is widely acclaimed for its many colors and innovative weaving techniques.

Pigments give paints and dyes their color. Ocher is a mineral-based pigment used to make several colors, including red. Red ocher gets its color from iron oxide. Pigments can also be plant-based; plant-based pigments contain a high level of carbon. In a 2023 study, archaeologists tested the red pigment on decorated beads made by members of the Natufian culture approximately 15,000 years ago. The test showed that the pigment found on several beads contained no iron but had a high level of carbon. This finding led the researchers to conclude that _____ blank

Which choice most logically completes the text?

- A. the Natufian beads examined in the study are the oldest surviving examples of the use of plant-based pigments for decorating beads.
- B. the Natufian beadmakers used plant-based pigments rather than ocher to decorate some of the beads examined in the study.
- C. the Natufian beadmakers preferred to use plant-based pigments because they are much brighter than mineral-based pigments are.
- D. the pigments used by the Natufian beadmakers likely came from plants because ocher was difficult to find.

The Land of Enchantment is a 1906 travel book by Lilian Whiting. In the book, which describes the experience of traveling through the southwestern United States by train, Whiting reflects on the escape from everyday life that such a journey provides: _____ blank

Which quotation from *The Land of Enchantment* most effectively illustrates the claim?

- A. “The opportunities and advantages already offered and constantly increasing are greater than would at first be considered possible.”
- B. “The social and the picturesque charm of the long journey is singularly enhanced by the leisurely stops made for refreshment.”
- C. “The real journey begins, of course, at Chicago, and as these trains leave in the evening the traveller fares forth in the seclusion of his berth.”
- D. “One experiences a certain sense of detachment from ordinary day and daylight duties that is exhilarating.”